

Internet Of Things

Open-Ended Lab



OEL

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Table of Contents

1	Introduction
2	Problem Analysis
3	Design Requirements
4	Possible Solutions
5	Preliminary Design
6	Design Description
7	Software Simulation
8	Experimental Results
9	Performance Analysis
10	Conclusion

References

Introduction

The main idea of this project is to design and implement a smart door automation system that automatically responds to environmental conditions using sensors and a microcontroller. The system monitors temperature and humidity values using sensors and controls an LED that represents the opening and closing of a smart door. If the surrounding temperature or humidity exceeds the defined threshold values, the LED turns ON to indicate that the door has opened for environmental regulation. If the readings remain below the threshold values, the LED stays OFF, representing a closed door. The project was implemented using the ESP8266 NodeMCU development board along with DHT11 and LM35 sensors.

The project mainly focuses on the concepts of data acquisition, sensor interfacing, automation, and actuation. Data acquisition refers to collecting real-time environmental data from sensors. Sensor interfacing is the process of connecting sensors with the microcontroller and reading their outputs through programming. Automation is achieved by applying conditional logic in the code so the system can make decisions without human involvement. Actuation in this project is represented by the LED, which acts as the output device simulating the movement of a smart door. These concepts are important because they form the foundation of modern embedded systems and Internet of Things (IoT) applications used in smart homes, industries, and environmental monitoring systems.

Initially, the LM35 temperature sensor was used to measure temperature through analog readings. However, unstable and inaccurate values were observed during implementation. Due to this issue, the DHT11 sensor was introduced as an improved solution because it can measure both temperature and humidity digitally with better stability. This introduced the concept of digital environmental sensing, which was not explored deeply in the course. The ESP8266 WiFi module was also used to connect the system to a wireless network and display connectivity information such as the IP address. This adds an IoT-related feature to the project and demonstrates how smart devices can communicate over networks in real-world automation systems.

Problem Analysis

The problem given in this project was to design and implement a smart door automation system that can monitor environmental conditions and respond automatically based on sensor readings. The project specifically required the use of temperature and humidity sensors to detect changes in the surrounding environment and then control an output device accordingly. Since an actual automated door mechanism was not available in the lab, an LED was used to simulate the opening and closing of the door. The system also needed to display real-time sensor readings on the Serial Monitor and allow threshold values to be defined for automation control.

To solve this problem, an ESP8266 NodeMCU development board was used as the main controller of the system. Initially, the LM35 temperature sensor was connected to measure room temperature using analog input readings. The sensor values were processed in the program, and if the temperature exceeded the defined threshold value, the LED would turn ON to represent the door opening. However, during implementation, the LM35 sensor produced unstable and inaccurate readings. Due to this issue, the DHT11 sensor was selected as an alternative solution because it provides more stable digital readings and can measure both temperature and humidity simultaneously.

The DHT11 sensor was interfaced with the ESP8266 using the DHT library in Arduino IDE. The program continuously reads temperature and humidity values and compares them with predefined threshold values. If the temperature becomes greater than 30°C or the humidity becomes greater than 70%, the LED connected to pin D2 turns ON, indicating that the smart door has opened for environmental regulation. Otherwise, the LED remains OFF. The sensor values and door status are displayed continuously on the Serial Monitor for monitoring and testing purposes. This approach successfully fulfilled the project requirements by combining sensing, processing, decision making, and output control in a simple automation system.

Design Requirements

The design requirements of this project focused on developing a smart door automation system that can monitor environmental conditions and respond automatically based on sensor readings. The system was required to measure temperature and humidity in real time and control an output device according to predefined threshold values. The project also required displaying sensor readings and system status on the Serial Monitor for monitoring and testing purposes.

The main inputs of the system are the temperature and humidity values obtained from the sensors. Initially, the LM35 sensor was used to measure temperature through analog input readings. However, due to unstable and inaccurate values during implementation, the DHT11 sensor was later used because it provides more reliable digital readings and can measure both temperature and humidity simultaneously. These sensor readings are continuously sent to the ESP8266 NodeMCU, which acts as the processing unit of the system. The microcontroller compares the received values with threshold conditions defined in the program. In this project, the temperature threshold was set to 30°C and the humidity threshold was set to 70%.

The output of the system is an LED connected to pin D2 of the ESP8266. Since an actual automatic door mechanism was not available, the LED was used to simulate the opening and closing of the door. When the temperature or humidity exceeds the threshold values, the LED turns ON to indicate that the door is open. Otherwise, the LED remains OFF. A major constraint during the project was sensor accuracy, which led to replacing the LM35 sensor with the DHT11 sensor for improved performance and stability.

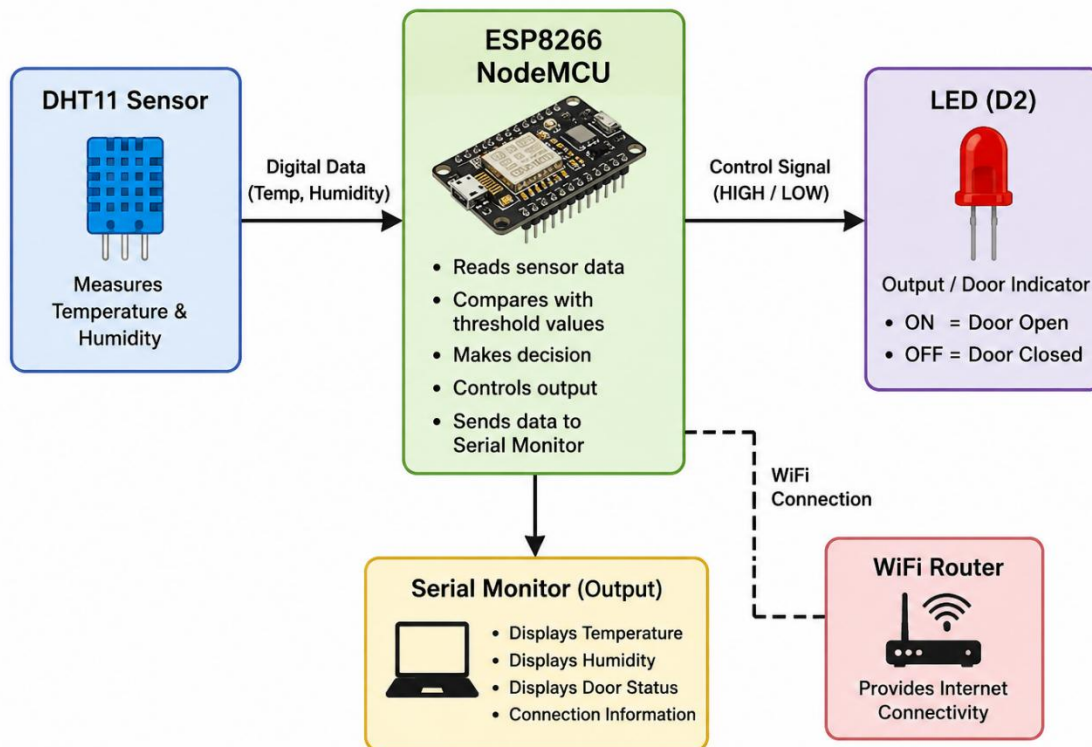
Possible Solutions

Several possible solutions were available for implementing the smart door automation system. One possible solution was to use the LM35 temperature sensor with the ESP8266 NodeMCU. The LM35 is a simple analog temperature sensor that provides voltage values proportional to temperature. In this method, the analog readings are converted into voltage values and then into temperature values through calculations in the program. The system could then compare the temperature with a threshold value and control the LED accordingly. This solution is simple and low cost, but during implementation the sensor produced unstable and inaccurate readings, which reduced the reliability of the system.

Another possible solution was to use the DHT11 sensor. Unlike the LM35, the DHT11 provides digital readings and can measure both temperature and humidity simultaneously. This allows the system to monitor multiple environmental conditions using a single sensor. The DHT11 also provides more stable readings and reduces errors caused by analog signal fluctuations. In this method, the ESP8266 reads temperature and humidity values directly through the DHT library and compares them with predefined threshold values to control the LED.

Between the two solutions, the DHT11-based system was selected as the final and best solution. The main reason for this choice was its better stability, dual sensing capability, and easier interfacing with the ESP8266. The LM35 only measures temperature and requires analog conversion, while the DHT11 provides direct digital values for both temperature and humidity. The DHT11 solution also improved system reliability and fulfilled the project requirements more effectively. Therefore, using the DHT11 sensor with ESP8266 was considered the most suitable solution for the smart door automation system.

Preliminary Design



Design Description

The smart door automation system consists of several important hardware and software blocks that work together to perform environmental monitoring and automatic control. The first block is the DHT11 sensor block. This sensor is responsible for measuring temperature and humidity from the surrounding environment. The DHT11 sends digital sensor readings directly to the ESP8266 NodeMCU. It was selected because it provides stable readings and can measure both parameters using a single sensor.

The second block is the ESP8266 NodeMCU microcontroller block, which acts as the main processing unit of the system. It receives sensor data from the DHT11 sensor and processes it according to the programmed conditions. The controller continuously compares the incoming temperature and humidity values with predefined threshold values. In this project, the temperature threshold was set to 30°C and the humidity threshold was set to 70%. The ESP8266 also manages WiFi connectivity and serial communication.

The third block is the LED output block. Since a real automated door mechanism was not available, an LED connected to pin D2 was used to simulate the opening and closing of the smart door. When the sensor readings exceed the threshold values, the ESP8266 sends a HIGH signal to the LED, turning it ON to represent an open door. If the readings remain below the thresholds, the LED stays OFF, representing a closed door.

Another important block is the Serial Monitor output block. This block displays real-time temperature, humidity, and door status readings for monitoring and debugging purposes. Finally, the WiFi block allows the ESP8266 to connect to a wireless network and display connection information such as the IP address, demonstrating basic IoT functionality in the system.

Software Simulation

```
// DHT11 CODE
#include <ESP8266WiFi.h>
#include <DHT.h>
// WiFi
const char* ssid = "m";
const char* password = "1122334455";
// Pins
#define DHTPIN D4
#define DHTTYPE DHT11
#define LEDPIN D2
DHT dht(DHTPIN, DHTTYPE);
// Thresholds
float tempThreshold = 30.0;
float humThreshold = 70.0;
void setup() {
  Serial.begin(115200);
  dht.begin();
  pinMode(LEDPIN, OUTPUT);
  digitalWrite(LEDPIN, LOW);
  // Connect to WiFi
  Serial.println();
  Serial.print("Connecting to WiFi: ");
  WiFi.begin(ssid, password);
  while (WiFi.status() != WL_CONNECTED) {
    delay(500);
    Serial.print(".");
  }
  Serial.println("\nConnected!");
  Serial.print("IP Address: ");
  Serial.println(WiFi.localIP());
}
void loop() {
  float temperature = dht.readTemperature();
  float humidity = dht.readHumidity();
```

```
// LM35 CODE
#include <ESP8266WiFi.h>
// WiFi Credentials
const char* ssid = "m";
const char* password = "1122334455";
// Pins
#define LM35PIN A0
#define LEDPIN D2
// Threshold
float tempThreshold = 30.0;
void setup() {
  Serial.begin(115200);
  pinMode(LEDPIN, OUTPUT);
  digitalWrite(LEDPIN, LOW);
  // Connect to WiFi
  Serial.println();
  Serial.print("Connecting to WiFi: ");
  WiFi.begin(ssid, password);
  while (WiFi.status() != WL_CONNECTED) {
    delay(500);
    Serial.print(".");
  }
  Serial.println("\nConnected!");
  Serial.print("IP Address: ");
  Serial.println(WiFi.localIP());
}
void loop() {
  int analogValue = analogRead(LM35PIN);
  float voltage = analogValue * (3.3 / 1023.0);
  float temperature = voltage * 100.0;
  // Print values
  Serial.print("Analog Value: ");
  Serial.print(analogValue);
  Serial.print(" | Voltage: ");
```

Software Simulation

```
// Check sensor
if (isnan(temperature) || isnan(humidity)) {
    Serial.println("Failed to read from DHT sensor!");
    delay(2000);
    return;
}

// Print readings
Serial.print("Temperature: ");
Serial.print(temperature);
Serial.print(" °C | Humidity: ");
Serial.print(humidity);
Serial.println(" %");

// Door Logic (LED)
if (temperature > tempThreshold || humidity > humThreshold) {
    digitalWrite(LEDPIN, HIGH);
    Serial.println("Door Status: OPEN (LED ON)");
} else {
    digitalWrite(LEDPIN, LOW);
    Serial.println("Door Status: CLOSED (LED OFF)");
}

Serial.println("-----");
delay(2000);
}
```

```
Serial.print(voltage);

Serial.print(" V | Temperature: ");
Serial.print(temperature);
Serial.println(" °C");

// Door Logic (LED)
if (temperature > tempThreshold) {
    digitalWrite(LEDPIN, HIGH);
    Serial.println("Door Status: OPEN (LED ON)");
} else {
    digitalWrite(LEDPIN, LOW);
    Serial.println("Door Status: CLOSED (LED OFF)");
}

Serial.println("-----");
delay(3000);
}
```

Experimental Results

DHT11

```
Output  Serial Monitor X
Message (Enter to send message to 'NodeMCU 1.0 (ESP-12E Module)' on 'COM3')
18:15:42.095 -> -----
18:15:44.092 -> Temperature: 30.20 °C | Humidity: 35.00 %
18:15:44.125 -> Door Status: OPEN (LED ON)
18:15:44.125 -> -----
18:15:46.113 -> Temperature: 30.20 °C | Humidity: 35.00 %
18:15:46.158 -> Door Status: OPEN (LED ON)
18:15:46.158 -> -----
18:15:48.174 -> Temperature: 29.80 °C | Humidity: 36.00 %
18:15:48.174 -> Door Status: CLOSED (LED OFF)
18:15:48.174 -> -----
18:15:50.172 -> Temperature: 29.80 °C | Humidity: 36.00 %
18:15:50.172 -> Door Status: CLOSED (LED OFF)
18:15:50.172 -> -----
18:15:52.226 -> Temperature: 29.80 °C | Humidity: 36.00 %
```

LM35

```
Output  Serial Monitor X
Message (Enter to send message to 'NodeMCU 1.0 (ESP-12E Module)' on 'COM3')
18:40:37.009 -> -----
18:40:40.018 -> Analog Value: 14 | Voltage: 0.05 V | Temperature: 4.52 °C
18:40:40.018 -> Door Status: CLOSED (LED OFF)
18:40:40.018 -> -----
18:40:43.005 -> Analog Value: 2 | Voltage: 0.01 V | Temperature: 0.65 °C
18:40:43.005 -> Door Status: CLOSED (LED OFF)
18:40:43.005 -> -----
18:40:46.015 -> Analog Value: 4 | Voltage: 0.01 V | Temperature: 1.29 °C
18:40:46.015 -> Door Status: CLOSED (LED OFF)
18:40:46.015 -> -----
18:40:48.981 -> Analog Value: 20 | Voltage: 0.06 V | Temperature: 6.45 °C
18:40:49.028 -> Door Status: CLOSED (LED OFF)
18:40:49.028 -> -----
18:40:51.996 -> Analog Value: 10 | Voltage: 0.03 V | Temperature: 3.23 °C
18:40:51.996 -> Door Status: CLOSED (LED OFF)
18:40:51.996 -> -----
18:40:54.983 -> Analog Value: 12 | Voltage: 0.04 V | Temperature: 3.87 °C
18:40:55.028 -> Door Status: CLOSED (LED OFF)
18:40:55.028 -> -----
18:40:57.994 -> Analog Value: 13 | Voltage: 0.04 V | Temperature: 4.19 °C
18:40:57.994 -> Door Status: CLOSED (LED OFF)
18:40:57.994 -> -----
```

Performance Analysis

The results obtained from the implementation showed that the smart door automation system worked successfully according to the required conditions. During testing, the DHT11 sensor was able to measure temperature and humidity values in real time and send the readings to the ESP8266 NodeMCU. The sensor values were displayed continuously on the Serial Monitor along with the door status. When the temperature exceeded the threshold value of 30°C or the humidity exceeded 70%, the LED turned ON, indicating that the smart door was open. When the values remained below the thresholds, the LED stayed OFF, representing a closed door. This confirmed that the automation logic and threshold comparison conditions were functioning correctly.

The performance of the DHT11-based system was stable and reliable throughout testing. The sensor provided consistent digital readings without major fluctuations, allowing the system to make accurate decisions. The response time of the system was also satisfactory, as the LED changed state almost immediately after the threshold conditions were met. The Serial Monitor output further confirmed correct operation by displaying updated environmental readings every few seconds.

Initially, the LM35 sensor was tested as a possible solution for temperature monitoring. Although the analog-to-temperature conversion logic was implemented correctly, the sensor produced unstable and unrealistic temperature readings during practical testing. Due to these inaccurate values, the LED remained OFF most of the time, reducing the effectiveness of the automation system. Because of this issue, the LM35 solution was replaced with the DHT11 sensor, which significantly improved the overall system performance.

Overall, the final DHT11-based system successfully achieved the objectives of environmental monitoring, automatic decision making, and output control in a simple and effective smart automation application.

Conclusion

In conclusion, the smart door automation system was successfully designed and implemented using the ESP8266 NodeMCU and DHT11 sensor. The system achieved the main objective of monitoring environmental conditions and automatically controlling an output device based on predefined threshold values. Temperature and humidity readings were continuously measured and displayed on the Serial Monitor, while the LED successfully simulated the opening and closing of a smart door. The automation logic worked correctly, as the LED turned ON whenever the temperature exceeded 30°C or the humidity exceeded 70%, and remained OFF under normal conditions.

During the development process, different solutions were explored to improve system performance. Initially, the LM35 sensor was tested for temperature monitoring, but it produced unstable and inaccurate readings during implementation. Due to this limitation, the DHT11 sensor was selected as the final solution because it provided more reliable digital readings and could measure both temperature and humidity simultaneously. This decision improved the accuracy, stability, and overall reliability of the system.

The project also demonstrated important concepts related to embedded systems, sensor interfacing, automation, and IoT-based applications. The ESP8266 NodeMCU effectively handled sensor data processing, decision making, WiFi connectivity, and output control within a single system. Despite hardware limitations, such as the absence of a real automatic door mechanism, the LED simulation successfully represented the required functionality.

Overall, the developed solution proved to be simple, cost-effective, and efficient for environmental monitoring and smart automation applications. The project successfully fulfilled all the required objectives and provided practical understanding of real-time sensing and automated control systems.

References
